

Proofs

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1 Proofs

1.1 Proof by contrapositive

Prove: if n^2 is even, then n is even

Let $n \in \mathbb{Z}$. Direct proof isn't possible, as it requires taking the square root of both sides of $n^2 = 2k$. Contrapositive proof means if n is odd, then n^2 must also be odd. $n = 2k + 1$ is odd, because $2k$ is even, as any number divisible by 2 is even $\frac{2k}{2} = k$, and any even number plus one is odd. To prove n^2 is odd:

$$\begin{aligned}n &= 2k + 1 \\n^2 &= (2k + 1)^2 \\&= 4k^2 + 4k + 1 \\n^2 &= 2(2k^2 + 2k) + 1\end{aligned}$$

Theorem: If n^2 is even, then n is even. Proof(contrapositive): If n is odd, then n^2 is odd. Let $n = 2k + 1$ for some integer k . Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. Hence n^2 forces n even. ■

Theorem: If n^2 is even, then n is even. (n^2 even $\implies n$ even, $n \in \mathbb{Z}$)

Proof (contrapositive): If n is odd, n^2 is odd. Let $n \in \mathbb{Z}$, then from the definition of an odd number, for some $k \in \mathbb{Z}$,

$$n = 2k + 1$$

Therefore,

$$\begin{aligned}n^2 &= (2k + 1)^2 \\&= 4k^2 + 4k + 1 \\&= 2(2k^2 + 2k) + 1\end{aligned}$$

Since $(2k^2 + 2k) \in \mathbb{Z}$, n^2 is odd. Thus the contrapositive is true, if n^2 is even, then n is even.

Theorem: n^2 even $\implies n$ even, $n \in \mathbb{Z}$, $\therefore \neg n$ even $\implies \neg n^2$ even, where $\neg n$ even $\iff n$ odd. $\therefore n$ odd $\implies n^2$ odd. ■

2 Questions

1. Spot the flaw: Claim $2 = 1$. Let $a = b$. Then $a^2 = ab$ so $a^2 - b^2 = ab - b^2$, so $(a+b)(a-b) = b(a-b)$. Dividing both sides by $a - b : a + b = b$. Since $a = b : 2b = b$ so $2 = 1$
The obvious flaw in this logic is the division by $a - b$, which while $a = b$ means that we are dividing by 0.
2. Theorem: If n^3 is even, n is even.
Proof (contrapositive): If n is odd, then n^3 is odd. Using the definition of an odd number, let $n = 2k + 1$, where $n, k \in \mathbb{Z}$.

$$\begin{aligned}n^3 &= (2k + 1)^3 \\&= (2k + 1)(2k + 1)(2k + 1) \\&= 8k^3 + 12k^2 + 6k + 1 \\&= 2(4k^3 + 6k^2 + 3k) + 1\end{aligned}$$

Plugging 1 into the equation,

$$2(4(1)^3 + 6(1)^2 + 3(1)) + 1 = 2(13) + 1 = 27$$

Since $(4k^3 + 6k^2 + 3) \in \mathbb{Z}$, n^3 is odd. Thus the contrapositive is true, if n^3 is even, n is even. ■

3. Theorem: The sum of two odd integers is even.
Proof (direct): Let $a, b \in 2\mathbb{Z} + 1$ By definition of odd numbers, $a = 2k + 1$ and $b = 2m + 1$ for some $k, m \in \mathbb{Z}$

$$\begin{aligned}a + b &= (2k + 1) + (2m + 1) \\&= 2k + 2m + 2 \\&= 2(k + m + 1)\end{aligned}$$

$k, m \in \mathbb{Z} \implies k + m + 1 \in \mathbb{Z}$, so $2(k + m + 1)$ must be even $\therefore a + b$ is even, so the sum of two odd numbers is even. ■

4. Theorem: There is no largest even integer.
Proof (contradiction): Suppose we have the largest $n \in 2\mathbb{Z}$. Since n is even, $n = 2k$ for some $k \in \mathbb{Z}$. Consider the integer $n + 2$:

$$n + 2 = 2k + 2 = 2(k + 1)$$

Since $k + 1 \in \mathbb{Z}$, $n + 2 \in 2\mathbb{Z}$. Furthermore, because $2 > 0$, we have $n + 2 > n$. This means that $n + 2 \in \mathbb{Z}$ and $n + 2 > n$, which contradicts our assumption that n is the largest $\in \mathbb{Z}$. $\therefore$ There is no largest even integer. ■

5. State the converse of $n > 2 \implies n^2 > 4$ and decide if it is true - proof or counterexample.
The converse is $n^2 > 4 \implies n > 2$.
Theorem: The statement $n^2 > 4 \implies n > 2$ is false. Proof (counterexample): Let $n \in \mathbb{R}$, suppose $n = -3$ then $(-3)^2 = 9$ which > 4 , so $n^2 > 4$. However, $-3 \leq 2$, so $n \leq 2$, $\therefore n^2 > 4 \not\implies n > 2$.
■

3 Format testing

1. **Theorem:** If n^3 is even, n is even

Proof (contrapositive): If n is odd, n^3 is odd. Let $n \in \mathbb{Z}$ By definition of odd numbers,

$$n = 2k + 1 \quad \text{for some } k \in \mathbb{Z}$$

So for n^3 ,

$$\begin{aligned} n^3 &= (2k + 1)^3 \\ &= (2k + 1)(2k + 1)(2k + 1) \\ &= 8k^3 + 12k^2 + 6k + 1 \\ &= 2(4k^3 + 6k^2 + 3k) + 1 \end{aligned}$$

Since $k \in \mathbb{Z}$ and integers are closed under addition and multiplication,

$$4k^3 + 6k^2 + 3k \in \mathbb{Z}$$

Let $m = 4k^3 + 6k^2 + 3k$. Then $m \in \mathbb{Z}$ and

$$n^3 = 2m + 1$$

By definition of an odd integer n is odd. n being odd $\implies n^3$ is odd. This proves the contrapositive, $n^3 \implies n$ ■

2. **Theorem:** The sum of two odd integers is even.

Proof (direct): Let $a, b \in \mathbb{Z}$ be odd. By definition of odd numbers,

$$a = 2k + 1 \quad \text{and} \quad b = 2m + 1$$

for some $k, m \in \mathbb{Z}$

$$\begin{aligned} a + b &= (2k + 1) + (2m + 1) \\ &= 2k + 2m + 2 \\ &= 2(k + m + 1) \end{aligned}$$

Since $k, m \in \mathbb{Z}$ and integers are closed under addition,

$$k + m + 1 \in \mathbb{Z}$$

Let $n = k + m + 1$. Then $n \in \mathbb{Z}$ and

$$a + b = 2n$$

By definition of an even integer, $a + b$ is even. $\therefore$ the sum of two odd integers is always even. ■

3. **Theorem:** There is no largest even integer.

Proof (contradiction): Suppose we have the largest even integer n . Since n is even,

$$n = 2k \text{ for some } k \in \mathbb{Z}$$

Consider the even integer $n + 2$,

$$n + 2 = 2k + 2 = 2(k + 1)$$

$k + 1 \in \mathbb{Z} \implies n + 2 \in \mathbb{Z}$. Furthermore, because $2 > 0$ then $n + 2 > n$.

So $n + 2$ is even, an integer and larger than n which contradicts n being the largest even integer.

$\therefore$ there is no largest even integer. ■

4. **Theorem:** The statement $n^2 > 4 \implies n > 2$ is false.

Proof (counterexample): Let $n \in \mathbb{R}$. Suppose,

$$n = -3$$

then

$$n^2 = (-3)^2 = 9$$

$9 > 4$ so $n^2 > 4$ is true. However, $-3 \leq 2$, so $n \leq 2$.

$$\therefore n^2 > 4 \not\implies n > 2 \quad \blacksquare$$